

Soapy Driver for the sBITX: A Software On-Ramp for the sBITX HF Transceiver

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overview of software defined radio

- when people hear sdr they think of usb radio dongles (rtl-sdr, sdr-play, etc)
- **sdr refers to a processing technique** used by many modern radios
 - flexradio, icom (7300 and later), xiegu, ...
- **compare to cd/dvd/blu-ray technology:**
 - on the media, everything is just ones and zeros
 - **hardware reads bits, software decodes bits**
 - the processing software has different algorithms for different data
 - music file vs movie file vs data backup etc
- **software defines the ways that the bits can be processed**
 - am, fm, sideband, rtty, ft8, sstv, atv, ...

sbitx photo (mouse for scale, credit funkamature)



sbitx is linux-friendly HF transceiver made in india

- web site: <https://www.hfsignals.com/index.php/sbitx-v3>
- is a 20 watt class software-defined hf transceiver
- has a **raspberry pi 4 inside**, along with radio hardware (filters, amps, etc)
- has a pi 5 upgrade developed by the community
- has a nice touchscreen (the pi 7 inch touch display)
- is a **good shape and size for pota**, runs from 12 volts
- its designer did bitx40, ubitx, etc (zbitx is raw and should be avoided)
- designer's goal is to make radios that are affordable for hams worldwide
- currently **costs \$429** plus shipping and (potentially enhanced) tariffs

problem: sbtx is like a phone with no app store

- **sbtx comes with only one radio app**
 - that one app is written specifically for the sbtx hardware
 - that one app doesn't work for any other hardware
- **also, no other app works on the sbtx hardware**
 - community has been enhancing the one app, but there are no others
- **the internal format of sbtx's radio data is not the one that other apps expect**
- yet this is an open source linux platform and other sdr apps exist so we should be able to fix this, right?

how do we open sbitx up to more apps?

- **create a software device driver** of a type that many sdr apps already know how to use
- this driver does the necessary data conversions internally
 - the apps don't even know about the unusual data format
- it also does other standard radio driver functionality
 - set frequency, switch to transmit mode, etc

sbitx's internal data format issue

- a basic way to look at the problem:
 - sbitx is mono (one signal channel)
 - most apps expect “stereo” (two signal channels)
- a more technical way:
 - sbitx hardware uses the “real” data format
 - **almost all sdr apps expect the “IQ” data format**
- luckily there are a few different mathematical ways to convert real to IQ and IQ to real
 - details out of scope for this talk...

the soapy sdr framework

- the **soapy sdr framework** supports many sdrs on many OSes
 - sdr applications use its applications programming interface
 - systems programmers create device drivers for each type of radio
 - almost all common sdrs are already supported (rtl-sdr, sdr-play, ...)
 - many sdr apps already support it (see list on next page)
 - the framework also provides useful command line utilities
- soapy sdr is cross-platform (linux, windows, mac-os)
- **we created a soapy sdr device driver for the sBITX**
 - as mentioned, this driver does format conversions and radio control

many different sdr apps now work on sbtx

- **sbitx**: original app, and a greatly enhanced 64-bit version
- **pihpsdr**: g0orx's original and dl1ycf's much updated version
- **deskhpsdr**: fork of dl1ycf aimed at desktop instead of touch screen
- **linhpsdr**: w4whl's fork of g0orx's original code, supports >1 RX stream
- **quisk**: simpler app whose source code is easy to learn (C, Python)
- **CubicSDR**: a more complex app with "dsp expert-frendly" code
- **SparkSDR**: has a leading-edge UI, supports ft8 internally
- **gqrx**: well-rounded, general-purpose sdr rx app
- **gnuradio**: key enabler for experimentation

status, availability, plans

- driver now works well enough to support our target apps
- **driver is still experimental and is undergoing many updates**
- files are private on github, will be public soon
- goal: driver and apps to be included in a future sbtx 64-bit image
- sbtx 64-bit team url: <https://github.com/drexjj/sbtx/wiki>

on to the demo!

- original sbitx app
- soapy sdr "find" command
- soapy sdr "info" command
- SparkSDR
- others?
- gnuradio (optional, to show the signal processing steps)